



AI Playbook and Toolkit for Public Health Departments

AI for Health Education and Learning Design

HED 100

This module helps public health staff use AI responsibly to support health education, workforce training, community education, and learning design. AI can help draft lesson outlines, create scenarios, adapt materials for different audiences, generate knowledge check questions, and review content for readability. However, AI-generated health education materials must be reviewed carefully because errors, inaccessible language, cultural mismatch, stigma, or outdated guidance can harm learners and communities. The module is not tied to any specific approved AI tool. Instead, it teaches learners to use only agency-approved tools, approved sources, and appropriate review processes when AI supports health education. It also emphasizes that educational content should remain grounded in public health evidence, community context, health literacy, language access, and accessibility.

Level	Introductory to Intermediate
Primary track	Communications and Health Education
Appears in tracks	health-education

Learning Objectives

- Describe appropriate and inappropriate uses of AI in health education and learning design.
- Use approved source materials to guide AI-supported drafting of educational content.
- Review AI-generated materials for accuracy, health literacy, accessibility, cultural appropriateness, and stigma.
- Create a practical review workflow for AI-supported health education materials.
- Explain why approved tools and human review remain necessary before educational materials are shared.

Module Overview

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Jurisdiction and Agency Policy Note

This national-level module is intended for training and planning purposes. It does not replace legal, privacy, procurement, civil rights, records, cybersecurity, Tribal consultation, labor relations, human resources, or agency policy review. Learners should confirm applicable requirements in their own jurisdiction and organization before implementing AI-supported workflows.

Why This Topic Matters for Public Health AI

Health education is a core public health function because it helps people understand risks, prevention strategies, available services, and decisions that affect their health. When AI is used to draft or adapt educational materials, it can increase speed and provide useful starting points. The same speed can also spread inaccurate information or produce content that does not fit the audience's literacy level, cultural context, language needs, disability access needs, or lived experience.

Public health education often serves communities that have experienced unequal access, mistrust, discrimination, or poor communication from institutions. AI-supported materials must therefore be reviewed for more than grammar and style. They should be reviewed for accuracy, plain language, accessibility, cultural humility, stigma, and whether the message supports informed action rather than confusion or blame.

Public Health Example

A health educator is creating a workshop for community health workers on heat illness prevention. The educator wants to use AI to generate a draft agenda, plain-language handout, role-play scenario, and five knowledge check questions. The content must be consistent with approved public health guidance, local emergency instructions, language access requirements, and the realities of workers who spend time outdoors.

A safe workflow begins with approved sources and a clear audience description. The educator uses AI only to draft from approved public materials, reviews the output for accuracy and readability, checks whether the examples avoid blame or stigma, and asks a community partner or bilingual reviewer to review adapted materials. The final materials

are not considered approved until they complete the agency's normal review process.

Technical, Programmatic, and Operational Deep Dive

AI can support health education by generating first drafts, restructuring content, suggesting examples, converting technical language into plain language, and creating practice questions. It can also help tailor materials for different audiences, such as parents, older adults, community health workers, school staff, clinicians, or emergency volunteers. These uses are helpful only when the AI is guided by accurate source material and reviewed by people with public health, instructional, and audience knowledge.

Effective AI-supported learning design should begin with learning objectives. Staff should identify what learners need to know, what they need to do, and what evidence will show that learning occurred. AI can help draft objectives, but educators should revise them so they are specific, measurable, and connected to public health practice.

Health education materials should also be reviewed for accessibility. This includes reading level, plain language, document structure, alt text for images, captioning for videos, color contrast, accessible tables, and compatibility with assistive technologies. AI can suggest accessibility improvements, but staff should still use established accessibility standards and local review processes.

Risks, Failure Modes, and Guardrails

A major risk is inaccurate or outdated health guidance. AI systems may produce plausible recommendations that conflict with current public health guidance, local policy, or program eligibility rules. Guardrails include using approved source documents, requiring subject matter expert review, documenting source materials, and prohibiting AI from inventing citations or recommendations.

Another risk is producing content that is technically correct but inappropriate for the audience. AI may generate examples that do not reflect local conditions, use stigmatizing language, assume access to resources that people do not have, or fail to address language and disability access needs. Guardrails include audience review, community partner input, readability testing, and accessibility review.

A third failure mode is using real learner, client, or participant information in prompts. Staff may be tempted to paste evaluation comments, participant questions, or case examples into AI tools to generate training content. Guardrails include using fictional or generalized examples, removing identifiers, following data classification rules, and using only approved tools for any sensitive information.

Application to AI-Supported Workflows

An AI-supported health education workflow should specify which materials can be used as source content, which AI tasks are allowed, who reviews the output, and how final approval occurs. For example, AI may be allowed to create a draft handout from approved guidance, but not to create final medical advice or change program eligibility language. The workflow should also clarify when communications, legal, clinical, program, or community review is required.

Educational workflows should preserve the educator's professional judgment. AI can accelerate drafting, but it cannot know the community context, local trust issues, learner needs, or implementation constraints unless those are provided and reviewed. The educator remains responsible for ensuring that materials are accurate, appropriate, actionable, and respectful.

Reflection Questions

Which parts of your health education or training work could AI support without using sensitive information?

What review steps would be needed before AI-supported educational materials could be used with the public?

How would you check whether AI-generated examples are culturally appropriate and accessible?

Practical Exercise

Select a public health education topic, such as heat safety, vaccination, lead prevention, emergency preparedness, or diabetes prevention. Draft an AI-supported learning design plan without using real participant data or sensitive information. The plan should include learning objectives, intended audience, approved source materials, proposed AI tasks, review steps, accessibility considerations, and final approval steps.

Then create one short example of AI-supported content, such as a draft agenda, role-play scenario, knowledge check, or plain-language handout outline. Clearly label the content as a draft and identify what review is needed before it could be used.

Example: For a heat safety workshop for outdoor workers, the learner identifies approved state and CDC guidance as source material, asks AI to draft a 30-minute agenda and role-play scenario, then reviews the output for local cooling center information, plain language, Spanish-language review needs, disability access, and stigma-free framing. The learner notes that final content requires program and communications review before use.

Expected Artifact or Evidence

AI-Supported Health Education Review Plan

Draft educational content example with review notes

Knowledge Check

- 1. What is the most important first step after completing the AI for Health Education and Learning Design module?
- 2. Which practice best supports responsible AI use in public health?
- 3. Which statement best reflects the risk or guardrail for this module: A major risk is inaccurate or outdated health guidance. AI systems may produce plausible recommendations that conflict with current public health guidance, local policy, or program eligibility rules. Guardrails include using approved source documents, requiring subject matter expert review, documenting source materials, and prohibiting AI from inventing citations or recommendations.
- Another risk is producing content that is technically correct but inappropriate for the audience. AI may generate examples that do not reflect local conditions, use stigmatizing language, assume access to resources that people do not have, or fail to address language and disability access needs. Guardrails include audience review, community partner input, readability testing, and accessibility review.
- A third failure mode is using real learner, client, or participant information in prompts. Staff may be tempted to paste evaluation comments, participant questions, or case examples into AI tools to generate training content. Guardrails include using fictional or generalized examples, removing identifiers, following data classification rules, and using only approved tools for any sensitive information.
- 4. When should a staff member escalate an AI-related concern?
- 5. What counts as stronger completion evidence for a training module?

References and Resources

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